Tangential Hinge Field

Cross-Version Equivalency Chart and Analog Map

How to read this chart
Rows identify one underlying structural idea. Columns show how that idea appeared in each historical version and where it resides in the final canonical model. “Retained” means unchanged in role; “narrowed” means given a stricter structural definition; “externalized” means moved to an optional runtime or architecture profile; “superseded” means no longer part of the canonical field.

1. Version Lineage

Version / phase	Characteristic form	Contribution	Canonical disposition
Discovery / projective genesis	0-sphere, NULL pair, CAR/CDR, CONS, centroid, projective mirror	Geometric source language and first finite coordinate forms	Historical foundation; only structural invariants carried forward
Projective CONS Runtime v0.1	Executable binary relation runtime; ordered CONS; XOR, masks, rotations, projection	Unified geometry and implementation path	Runtime profile, not the Tangential Hinge Field core
Projective CONS Runtime v0.2	Refined coordinate indexing, practical uses, explicit non-claims	Separated reduced indices from complete relations	Authority discipline retained as external implementation guidance
Projective CONS Runtime v0.3	Integrated OMINO/OMICRON/OMNICRON, Polybius, Metatron, Tetragrammatron, clocks/rulers	Most complete executable integration draft	Supplies analogs and optional bindings; not the final geometry
Block Runtime Resolution Model	NULL, BIND, XOR, META, ESC, FS, GS, RS, US; bounded non-mutating resolution	Locked runtime vocabulary and five-place scope forms	Runtime semantics externalized from field geometry
BIND–XOR / BIND–SWAP experiments	ESC, STAY, NEXT, PORT, MASK, SWAP vocabulary substitutions	Explored stream-control wording	Superseded; demonstrated that flow words were not the field
Four-page / five-page field experiments	state A/B, capacity, order, weight, scalar, legend	Discovered aligned pages and mirrored configuration	Collapsed into four mirrored plane classes plus zero hinge
Stateless–stated–stateful field	negative C/O/W/S, central state, positive C/O/W/S	Established symmetric nine-plane arrangement	Refined by splitting zero into P0:T0
Parity/source zero-pair field	[S− W− O− C− | P0:T0 | C+ O+ W+ S+]	Removed state as ordinary plane; source/parity became hinge	Retained exactly
Canonical Tangential Hinge Field v1.0	Four-place scope descriptions, dual zero hinge, Blackboard, alignment, adjudication	Self-contained structural geometry	Current canonical lock

2. Core Vocabulary Equivalency

Underlying role	Projective runtime	Block runtime	Experimental wording	Canonical field	Disposition	Final definition
Unbound coordinate	NULL	NULL	NULL	NULL	Retained	Explicit unbound place; not hidden nil
Directly established place	BIND as boundary/encapsulation	BIND	BIND	BIND	Narrowed	A directly established coordinate place
Difference relation	XOR	XOR	SWAP proposed as renamed XOR	XOR for pairwise deltas	Retained outside entry grammar	Equal-width symmetric difference
Coordinate selector	MASK / META	META	MASK	Not required by FIELD	Externalized	Runtime/profile concern
Context exit	ESC	ESC	STAY/NEXT/PORT experiments	Not required by FIELD	Externalized / superseded	Runtime continuation concern
Four ordered scopes	FS, GS, RS, US	FS > GS > RS > US	Same words, varying fifth terminal	FS, GS, RS, US	Retained	Only admissible entry coordinates
Completed field	CONS / US closure	US = all BIND plus ESC	US row with terminal word	US = [BIND,BIND,BIND,BIND]	Narrowed	Completed four-place scope description
Central origin	0x00 centroid / NULL pair	NULL / 0x00	state / legend / zero	P0:T0	Refined	Dual zero hinge: parity/source

3. Scope-Form Equivalency

Scope	Five-place runtime form	Flow-word experiments	Canonical four-place form	Invariant count	Analog
FS	(BIND GS RS US ESC)	(BIND GS RS US NEXT / STAY / PORT)	[BIND, GS, RS, US]	1 direct + 3 subordinate	Outermost description of US
GS	(BIND RS US BIND ESC)	(BIND RS US BIND NEXT / STAY / PORT)	[BIND, RS, US, BIND]	2 direct + 2 subordinate	Intermediate description of US
RS	(BIND US BIND BIND ESC)	(BIND US BIND BIND NEXT / STAY / PORT)	[BIND, US, BIND, BIND]	3 direct + 1 subordinate	Immediate description of US
US	(BIND BIND BIND BIND ESC)	(BIND BIND BIND BIND NEXT / STAY / PORT)	[BIND, BIND, BIND, BIND]	4 direct + 0 subordinate	Completed field

Exact equivalence
The first four places never changed. The fifth word changed repeatedly because it belonged to a runtime continuation model, not to the scope geometry. Removing the fifth place reveals the stable invariant.

4. Plane-Model Equivalency

Concept	Earlier representation	Earlier role	Canonical analog	Relationship
Two active states	state A | state B	Two equally active data pages	Mirrored negative/positive references	Replaced by explicit paired planes C−↔C+, O−↔O+, W−↔W+, S−↔S+
Capacity	full byte plane or high nibble C	available extent / configuration class	C− and C+ at magnitude 1	Retained as nearest outer pair to hinge
Order	full byte plane or nibble O	place order / relation order	O− and O+ at magnitude 2	Retained; letter O distinguished from zero 0
Weight	full byte plane or nibble W	occupancy / weight class	W− and W+ at magnitude 3	Retained
Scalar	fifth scalar plane or nibble S	grade applied to capacity/weight/etc.	S− and S+ at magnitude 4	Retained as outermost pair
Legend	preheader or PLANE −1	one admissible reference configuration	P0 parity reference / external legend	Split: parity at hinge; semantic legend external
State	PLANE 0	central stated coordinate	T0 tangential source	Refined: source, not an ordinary measured plane
Symmetric field	S− W− O− C− | state | C+ O+ W+ S+	stateless–stated–stateful	S− W− O− C− | P0:T0 | C+ O+ W+ S+	Central single state split into oriented zero pair

5. Geometry and Addressing Analogs

Underlying geometry	Earlier name/form	Intermediate analog	Canonical name/form	Equivalence
16×16 byte board	Hexadecimal/ASCII coordinate plane	COBS–CONS Blackboard	16×16 Blackboard	Same 0x00…0xFF address surface
Row/column	0xNR positional lens	row=high nibble, col=low nibble	ADDRESS=0xRC	Same positional rule
Low/high charts	low 0x00…0x7F; high 0x80…0xFF	local/remote macro-surfaces	LOCAL/REMOTE	Same bit-7 partition
Projection	high(x)=x XOR 0x80	REMOTE=LOCAL XOR 0x80	same	Retained exact involution
32-position unit	bounding rectangle / lens	4 rows × 8 columns	tangential bounding lens	Retained
Eight units	4 local + 4 remote rectangles	8 × 32 = 256	complete Blackboard coverage	Retained
Low row endpoints	0x20…0x70 and 0x2F…0x7F	origin/control gauges	X0…XF	Retained as optional gauge profile
High row endpoints	0xA0…0xF0 and 0xAF…0xFF	projective origin/control gauges	X0…XF	Retained as optional gauge profile

6. Validation and Scale Analogs

Structure	Historical form	Numeric analog	Canonical placement	Meaning
Four scope dimensions	FS, GS, RS, US	4! = 24 orderings	24-position body	Scope-order resolution
Complete terminal	0x18…0x1F	8 positions	24+8 lens view	Complete COBS–CONS terminal profile
Operational gauge	A…F terminal	6 positions	26+6 lens view	OMINO operational profile
Tetragrammatron governor	0x1B…0x1F	5 positions	27+5 lens view	Byte/page relation jurisdiction
Metatron field	0x1C…0x1F	4 positions	28+4 lens view	Nibble/place incidence jurisdiction
Metatron radix	16^0…16^4	0x00001…0x10000	16^4 domain	Four-nibble / 16-bit adjudication
Tetragrammatron radix	256^0…256^4	0x000000001…0x100000000	256^4 domain	Four-byte / 32-bit adjudication
Azimuth	0xAA55	A|A|5|5 and AA|55	optional orientation profile	Bridge between nibble and byte readings
Clock	4320	when / synchrony	external timing coordinate c	Not part of FIELD geometry
Rotation ruler	5040 = 7!	which ordering	external rotation coordinate r	Not part of FIELD geometry

7. Artifact and Surface Analogs

Layer	Historical name	Role	Closest field analog	Boundary
User surface	.omi	user declaration	Optional architecture binding	Not part of FIELD core
System surface	.imo	system definition	Optional architecture binding	Not part of FIELD core
Machine surface	.o	machine coordinate/executable substrate	Optional architecture binding	Not part of FIELD core
Local half	CAR	carried/user-local coordinate	LOCAL Blackboard surface	Analog only; not canonical identity
Remote half	CDR	continuation/user-remote coordinate	REMOTE Blackboard surface	Analog only; not canonical identity
Complete relation	CONS	ordered CAR/CDR relation	FIELD / complete US attachment	Structural analogy; runtime object external
Five semantic families	RULES, FACTS, CLOSURES, COMBINATORS, CONS	15 concentric artifacts	No direct FIELD planes	Separate repository/artifact architecture

8. Retained, Narrowed, Externalized, and Superseded

Disposition	Items
Retained	FIELD mirror symmetry; P0:T0 hinge; FS/GS/RS/US ordering; 16×16 Blackboard; XOR 0x80 mirror; pairwise deltas; Metatron/Tetragrammatron radix ladders
Narrowed	BIND → directly established place; US → completed four-place scope; state → T0 tangential source; legend → P0 parity plus optional external legend
Externalized	XOR as runtime primitive; META; ESC; CAR/CDR; CONS serialization; OMINO/OMICRON/OMNICRON profiles; 0xAA55 orientation; clocks and slide rulers; .omi/.imo/.o artifact surfaces
Superseded	STAY/NEXT/PORT flow vocabulary; SWAP as a replacement operation name; five-place field entry grammar; state A/state B as the final model; interpreting parentheses as Lisp evaluation

9. One-Line Analog Map

Complete evolution
Ordered CONS runtime → bounded BIND/XOR scope runtime → flow-word experiments → aligned page model → symmetric state field → parity/source zero hinge → four-place FS–GS–RS–US entry geometry.

Earlier expression	Canonical equivalent
CONS cell / bounded relation	A complete FIELD reference at one Blackboard address
CAR : CDR	Negative : positive orientation across the hinge (analogy, not identity)
NULL : NULL	P0:T0 zero-oriented hinge ancestry
FS > GS > RS > US	FS, GS, RS are ordered descriptions of completed US
Five-place scope + ESC	Four-place scope; ESC belongs to runtime
state A XOR state B	paired deltas ΔC, ΔO, ΔW, ΔS
legend XOR state	P0 parity reference relative to T0 source
four active pages C/O/W/S	one positive quadrant Q+ = [C+ O+ W+ S+]
mirrored four pages	Q− | P0:T0 | Q+
0x00…0xFF byte plane	16×16 Blackboard address domain

Earlier expression	Canonical equivalent
low XOR 0x80 = high	LOCAL XOR 0x80 = REMOTE
Metatron / Tetragrammatron	nested incidence / relation adjudication

10. Final Canonical Crosswalk

Canonical element	Historical ancestors / analogs	Do not confuse with
FIELD = [S− W− O− C− | P0:T0 | C+ O+ W+ S+]	four-page configuration; five-plane scalar model; nine-plane state field; mirrored Blackboard pages	an executable list or stored record format
P0:T0	NULL pair; legend/state comparison; parity/source zero pair; tangential pinch/hinge	two different numeric zeros
FS/GS/RS/US	tetrahedral bounds; Hamming data coordinates; four-scope permutation kernel	four semantic domains or four independent definitions
BIND	encapsulation; bounded relation; known coordinate	AND, pointer assignment, function application, or mutation
ADDRESS = 0xRC	hex byte coordinate; Polybius-like row/column lens	a 16-bit 0xRRCC address unless a wider table profile is declared
ΔC, ΔO, ΔW, ΔS	XOR witnesses; alignment surface; STDERR-like distinction analogy	one combined XOR proof
Metatron	four-nibble place anchors; 28+4 field	Tetragrammatron byte-scale relation
Tetragrammatron	four-byte anchors; 27+5 governor	Metatron nibble-scale incidence

Canonical reading rule
When comparing any older document to the final field, first ask whether the term describes geometry, runtime behavior, timing, representation, or an optional profile. Only geometric invariants belong in the Tangential Hinge Field core.